Task 1
Answer Sheet

Waters of Croatia: From the Emerald Rivers to the Deep Blue Sea

EOES2025 Zagreb Croatia
26.04. – 03.05.2025

Team ___________________

Students ____

____

____

European Olympiad of Experimental Science

Problem 1. Ready, Set, Go!

(TOTAL MARKS: 47)

Step 1.1.

*1.1.1. Fill in Table 1.1.1. following the instructions. Count the organisms in Sample A and Sample B, and write the names of the taxonomic groups and the number of individuals in each sample in each taxonomic group in this table. Calculate the benthic density (number of individuals/m^2) for each taxonomic group, based on the sample area of 0.049 m^2. Round the calculation of benthic density to two decimal places. Also fill in the columns where you should check if your organisms are segmented (i.e., if they have distinct body segments along the longitudinal axis) and the column where you should check if the organisms have legs. Indicate this by placing a tick (✓) in the appropriate column for organisms with these characteristics, or a slash (/) if they do not. (**Please note that the number of rows in this table does not indicate the number of individual taxa in each sample.**)* (19.25 p)

Table 1.1.1. Abundance and characteristics of organisms found in Samples A and B.

Sample	**Taxonomic group**	**Number of individuals in the sample**	**Benthic density (Number of individuals / m^2)**	**Morphology**		Marks
				Segmented	**Legs**	
Sample A						
Sample B						
					TOTAL MARKS	

1.1.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: ______	

1.1.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: ______	

Step 1.2.

1.2.1. Enter the calculated Simpson Diversity Index (SDI) (rounded to three decimal places) for your two samples (Sample C and Sample D) in the blanks provided and write the letter corresponding to the correct answer in the space provided. (3 p)

	Marks
SDI (Sample C): ____________________	
SDI (Sample D): ____________________	
Sample ___________ (write C or D in the space provided) has greater diversity.	
TOTAL MARKS	

1.2.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: ______	

1.2.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: ______	

Step 1.3.

1.3.1. Enter the calculated proportions of individuals belonging to mayflies (Ephemeroptera), stoneflies (Plecoptera), and caddisflies (Trichoptera) in your two samples (Sample C and Sample D), expressed as percentages (%), rounded to one decimal place, in Table 1.3.1. below. (4 p)

Table 1.3.1. Proportions (%) of mayflies (Ephemeroptera), stoneflies (Plecoptera), and caddisflies (Trichoptera) in Samples C and D.

	Sample C	**Sample D**	Marks
E; Ephemeroptera (%)			
P; Plecoptera (%)			
T; Trichoptera (%)			
EPT; Ephemeroptera + Plecoptera + Trichoptera (%)			
		TOTAL MARKS	

1.3.2. Enter the calculated proportions of individuals belonging to Diptera and Oligochaeta in your two samples (Sample C and Sample D), expressed as percentages (%), rounded to one decimal place, in Table 1.3.2. below. (3 p)

Table 1.3.2. Proportions (%) of Diptera and Oligochaeta in Samples C and D.

	Sample C	**Sample D**	Marks
D; Diptera (%)			
O; Oligochaeta (%)			
DO; Diptera + Oligochaeta (%)			
		TOTAL MARKS	

1.3.3. Please enter the letter of the correct answer in the box. (0.5 p)

	Marks
Correct answer: _______	

Step 1.4.

1.4.1. Enter your answers in Table 1.4.1. below by placing a tick (✓) in the appropriate column if the organisms have the listed characteristics or a slash (/) if this is not the case. (2.75 p)

Table 1.4.1. Characteristics (feeding types) of organisms found in Samples C and D.

Sample	**Taxonomic group**	**Feeding types**		Marks
		Herbivorous/ Detritivorous	**Predators**	
Sample C	Isopoda			
	Ephemeroptera			
	Diptera			
	Oligochaeta			
	Bivalvia			
	Gastropoda			
Sample D	Amphipoda			
	Ephemeroptera			
	Plecoptera			
	Trichoptera			
	Diptera			
			TOTAL MARKS	

1.4.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: _______	

1.4.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: _______	

1.4.4. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: _______	

Step 1.5.

1.5.1. Enter the results of your calculations (rounded to two decimal places) for the volume of water passing through each drift sampler during the sampling period, the drift density (number of organisms/m^3) for each taxonomic group in each drift sample, and the averages for all parameters in Table 1.5.1. below. (8 p)

Table 1.5.1. Results of drift calculations based on the data obtained during and after the collection of the drift samples.

	Volume of water filtered through the net	**Ephemeroptera drift density (Number of organisms/m^3)**	**Amphipoda drift density (Number of organisms/m^3)**	**Diptera drift density (Number of organisms/m^3)**	Marks
Drift net 1					
Drift net 2					
Drift net 3					
AVERAGE					
				TOTAL MARKS	

1.5.2. Enter the results of your calculations (rounded to three decimal places) for the drift propensity for each taxonomic group in the Table 1.5.2. below. (1.5 p)

Table 1.5.2. Results of drift propensity calculations based on benthic and drift density data

	Benthic density (Number of individuals/m²)	**Drift density (Number of individuals/m³)**	**Drift propensity**	Marks
Ephemeroptera	429	10		
Amphipoda	3531	24		
Diptera	122	12		
			TOTAL MARKS	

1.5.3 Based on your results, draw conclusions about which groups are more likely to drift downstream. Enter your answers in Table 1.5.3. ***Use the symbols >, <, or = to indicate the relationships between the taxonomic groups in terms of their drift propensity*** (1.5 p)*:*

Table 1.5.3. Relationships between the taxonomic groups in terms of their drift propensity.

	Marks
Ephemeroptera _______ Diptera Amphipoda _______ Ephemeroptera Diptera _______ Amphipoda	

Problem 2. Salts from Croatian Saltworks

(TOTAL MARKS: 37)

Step 2.1. Chemical reactions for determination of iodine species

2.1.1. Fill the Table 2.1.1. (Answer Sheet) with your results by marking the observed change with an X. (3.5 p)

Table 2.1.1. Observations

Test tube		dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T1								
T2								
T3								
T4	a)							
	b)							
T5	a)							
	b)							
							TOTAL MARKS	

2.1.2. Write the balanced chemical reactions in test tubes T4a and T5a (write states of matter) into Table 2.1.2. (Answer Sheet). (1 p)

Table 2.1.2. Balanced chemical reactions in T4 and T5

Test tube		Balanced chemical reactions	Marks
T4	a)		
T5	a)		
		TOTAL MARKS	

Step 2.2. Testing salt solutions

2.2.1. Fill in the Table 2.2.1. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.1. Observed changes in sample Salt **1**

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T6							
T7							
T8							
						TOTAL MARKS	

2.2.2. Fill in the Table 2.2.2. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.2. Observed changes in sample Salt **2**

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T9							
T10							
T11							
						TOTAL MARKS	

2.2.3. Fill in the Table 2.2.3. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.3. Observed changes in sample Salt **3**

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T12							
T13							
T14							
						TOTAL MARKS	

2.2.4. In Table 2.2.4. mark with an X the correct form of iodine that is present in table salt samples or mark no iodine if it is not present. (1.5 p)

Table 2.2.4. Form of iodine present in the samples Salt **1**, Salt **2** and Salt **3**

Salt sample	I_2	I^-	IO_3^-	No iodine	Marks
Salt 1					
Salt 2					
Salt 3					
		TOTAL MARKS			

Confirmation of results. Signature of lab assistant: ______________________

Step 2.3. Measuring iodine in table salts

2.3.1. Determine the oxidation states of iodine in iodine molecule, iodate ion and iodide ion. Write the answer in the Table 2.3.1. on the Answer Sheet. (1.5 p)

Table 2.3.1. Oxidation states of iodine in iodine molecule (I_2), iodate ion (IO_3^-) and iodide ion (I^-)

	Oxidation state of iodine	**Marks**
Iodine (I_2)		
Iodate (IO_3^-)		
Iodide (I^-)		
	TOTAL MARKS	

2.3.2. In the reaction of iodine and thiosulfate ion, determine what species is being reduced and what species is being oxidized. Indicate your answer by X in appropriate cell in Table 2.3.2. on the Answer Sheet. (1 p)

$$I_2\,(aq) + 2S_2O_3^{2-}\,(aq) \rightarrow 2I^-\,(aq) + S_4O_6^{2-}\,(aq)$$

Table 2.3.2. Species being reduced and species being oxidized in the reaction of iodine and thiosulfate ion

	I_2	**$S_2O_3^{2-}$**	**Marks**
Species being reduced			
Species being oxidized			
		TOTAL MARKS	

2.3.3. What is the role of starch solution in the titration of iodine with the standard solution of sodium thiosulfate? Enter the corresponding letter in Answer Sheet. (0.5 p)

a) starch is used as indicator in the detection of endpoint of reaction involving iodine solution.

b) starch is used to stabilize the solution of iodine.

c) starch is used to achieve acid medium needed for the reduction of iodate ions into iodine.

Table 2.3.3. *Please write the letter corresponding to the correct answer in the space provided.*

	Marks
Correct answer: __ ___	

2.3.4. In the Table 2.3.4. mark with an X which salt samples have you chosen based on the previous results. (1 p)

Table 2.3.4. Samples chosen for quantitative analysis

Salt sample	Samples chosen for analysis	Marks
Salt 1		
Salt 2		
Salt 3		
	TOTAL MARKS	

Confirmation of results. Signature of lab assistant: ______________________

2.3.5. In the Table 2.3.5. ***write which procedure you have chosen (A, B, C or none), and raise the red card so the lab assistant can check your answer. Only after that can you start with the experiment. The lab assistant will check answers for tasks 2.2.4, 2.3.4. and 2.3.5. and sign them if the answers are correct. If the answers are incorrect, you will not get points for these tasks, but the lab assistant will provide you with correct answers.*** (1.5 p)

Table 2.3.5. Sample(s) and procedure(s) for quantitative analysis of salt samples

	Procedure	Marks
Salt 1		
Salt 2		
Salt 3		
	TOTAL MARKS	

Confirmation of results. Signature of lab assistant: ______________________

Correct answers

Salt 1 **Iodine species: ____________ ; procedure _________________** **Salt 2** **Iodine species: ____________ ; procedure _________________** **Salt 3** **Iodine species: ____________ ; procedure _________________** **Signature of lab assistant: ______________________**

(Total marks for 2.3.6 + 2.3.7 + 2.3.8: 12 p)

*2.3.6. If you analyzed sample Salt **1**, in Table 2.3.6. on the Answer sheet write the volumes of $Na_2S_2O_3$ used for titration, calculate the mean volume and circle the volumes used in your calculation.*

*If you didn't analyze sample Salt **1** write "**/**" in the cells of Table 2.3.6.*

Write down the concentration of $Na_2S_2O_3$ as indicated on the bottle _______________

Table 2.3.6. Volumes of $Na_2S_2O_3$ used for titration of the salt sample

Titration number	***V*($Na_2S_2O_3$) (mL)**	**Mean value $\bar{V}$($Na_2S_2O_3$) (mL)**	**Marks**
		TOTAL MARKS	

*2.3.7. If you analyzed sample Salt **2**, in Table 2.3.7. on the Answer sheet write the volumes of $Na_2S_2O_3$ used for titration, calculate the mean volume and circle the volumes used in your calculation.*
*If you didn't analyze sample Salt **2** write "**/**" in the cells of Table 2.3.7.*

Write down the concentration of $Na_2S_2O_3$ as indicated on the bottle _______________

Table 2.3.7. Volumes of $Na_2S_2O_3$ used for titration of the salt sample

Titration number	***V*($Na_2S_2O_3$) (mL)**	**Mean value $\bar{V}$($Na_2S_2O_3$) (mL)**	**Marks**
		TOTAL MARKS	

*2.3.8. If you analyzed sample Salt **3**, in Table 2.3.8. on the Answer sheet write the volumes of $Na_2S_2O_3$ used for for titration, , calculate the mean volume and circle the volumes used in your calculation.*

*If you didn't analyze sample Salt **3** write "**/**" in the cells of Table 2.3.8.*

Write down the concentration of $Na_2S_2O_3$ as indicated on the bottle _______________

Table 2.3.8. Volumes of $Na_2S_2O_3$ used for titration of the salt sample

Titration number	$V(Na_2S_2O_3)$ **(mL)**	**Mean value** $\bar{V}(Na_2S_2O_3)$ **(mL)**	**Marks**
		TOTAL MARKS	

2.3.9. Calculate the molar mass (in g mol^{-1} units) of sodium thiosulfate ($Na_2S_2O_3$), iodine molecule (I_2), potassium iodate (KIO_3) and potassium iodide (KI). Write the answer in the Table 2.3.9 (Answer sheet). (2 p)

Calculation

Table 2.3.9. Molar mass of sodium thiosulfate, iodine molecule, potassium iodate and potassium iodide

	***M* / g mol^{-1}**	**Marks**
$Na_2S_2O_3$		
I_2		
KIO_3		
KI		
	TOTAL MARKS	

*2.3.10. In the table 2.3.10. on the Answer sheet calculate the amount (in moles) of sodium thiosulfate used for titration of samples. Indicate the sample you are performing calculation for (sample Salt **1**, Salt **2** or Salt **3**).* (1 p)

Table 2.3.10. Calculation of moles of sodium thiosulfate used for titration of samples

	Marks
Sample: _____	
Sample: ______	
Sample: ________	
TOTAL MARKS	

*2.3.11. In the table 2.3.11. on the Answer sheet calculate the amount (in moles) of iodine species (I^-, I_2 or IO_3^-, according to the form of iodine which is present in the given sample) present in the initial sample solution (250 mL). Indicate the sample you are performing calculation for (sample Salt **1**, Salt **2** or Salt **3**).* (2 p)

Table 2.3.11. Calculation of moles of iodine species present in the sample(s)

	Marks
Sample: ______	
Sample: ______	
Sample: ________	
TOTAL MARKS	

*2.3.12. In the table 2.3.12. on the Answer sheet calculate the mass of iodine species (KI, I_2 or KIO_3, according to the form of iodine which is present in the given sample) in the initial sample solution (250 mL). Indicate the sample you are performing calculation for (sample Salt **1**, Salt **2** or Salt **3**).* (1 p)

Table 2.3.12. Calculation of the mass of iodine species present in the sample(s)

	Marks
Sample: ______	
Sample: ______	
Sample: ________	
TOTAL MARKS	

*2.3.13. In the table 2.3.13. on the Answer sheet calculate the fraction(as mg per kg of salt) of iodine species (KI, I_2 or KIO_3, according to the form of iodine which is present in the given sample) in the sample(s), rounded to 1 decimal place. Indicate the sample you are performing calculation for (sample Salt **1**, Salt **2** or Salt **3**).* (1 p)

Table 2.3.13. Calculation of the amount of iodine species in the sample(s) as mg of iodine species per kg of salt

	Marks
Sample: _____	
Sample: _______	
Sample: ________	
TOTAL MARKS	

*2.3.14. Based on the results, mark on the map the saltworks from which the salt samples (Salt **1**, Salt **2** and Salt **3**) came, if you know that the saltwork in Ston does not add iodine into their salt, and that saltwork in Pag adds more iodine into their salt than the saltwork in Nin.* (2 p)

	Marks
	

Problem 3. Bending Light in Aqueous NaCl Solution

(TOTAL MARKS: 36)

3.1.1. In Figure 3.2 a) and b) the incoming beam (gray dashed line) reaches the boundary between two media (full black line), in this case between sapphire and ethanol. (Two protractors are also drawn on the Figure, one on each side of a border between two media.) Based on the data from Table 3.1 and Snell's law, draw the refracted rays for both cases on the Answer Sheet. (The accuracy of the drawing may not be worse than ±1 degree.) Show your calculation. (2 p + 2 p)

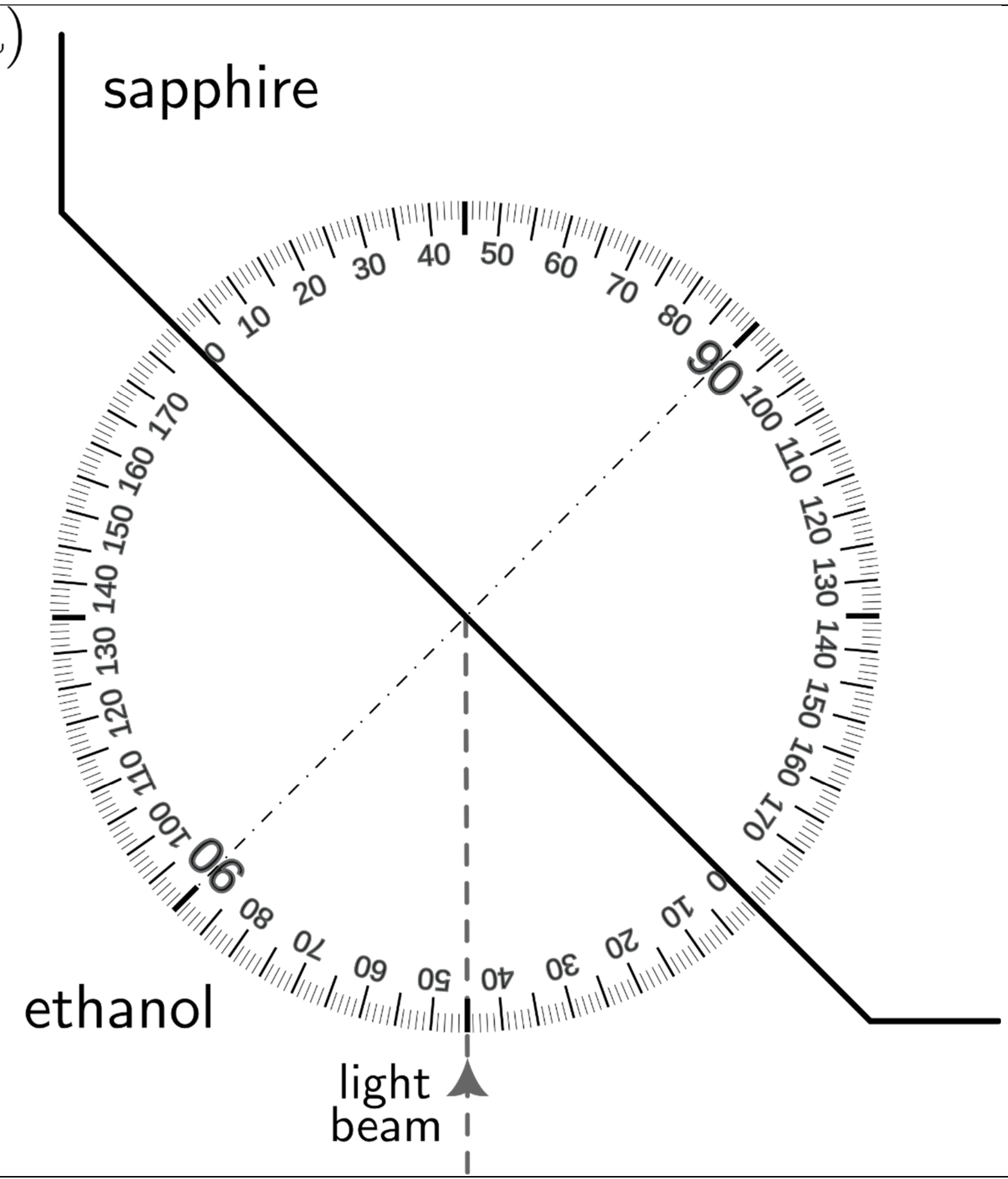
a)	Marks
	
Calculation	

Figure 3.2. Full black line: the boundary between ethanol and sapphire. Gray dashed line: the incoming light beam (the arrow indicates the direction of the propagation). a) The beam reaches the boundary from ethanol.

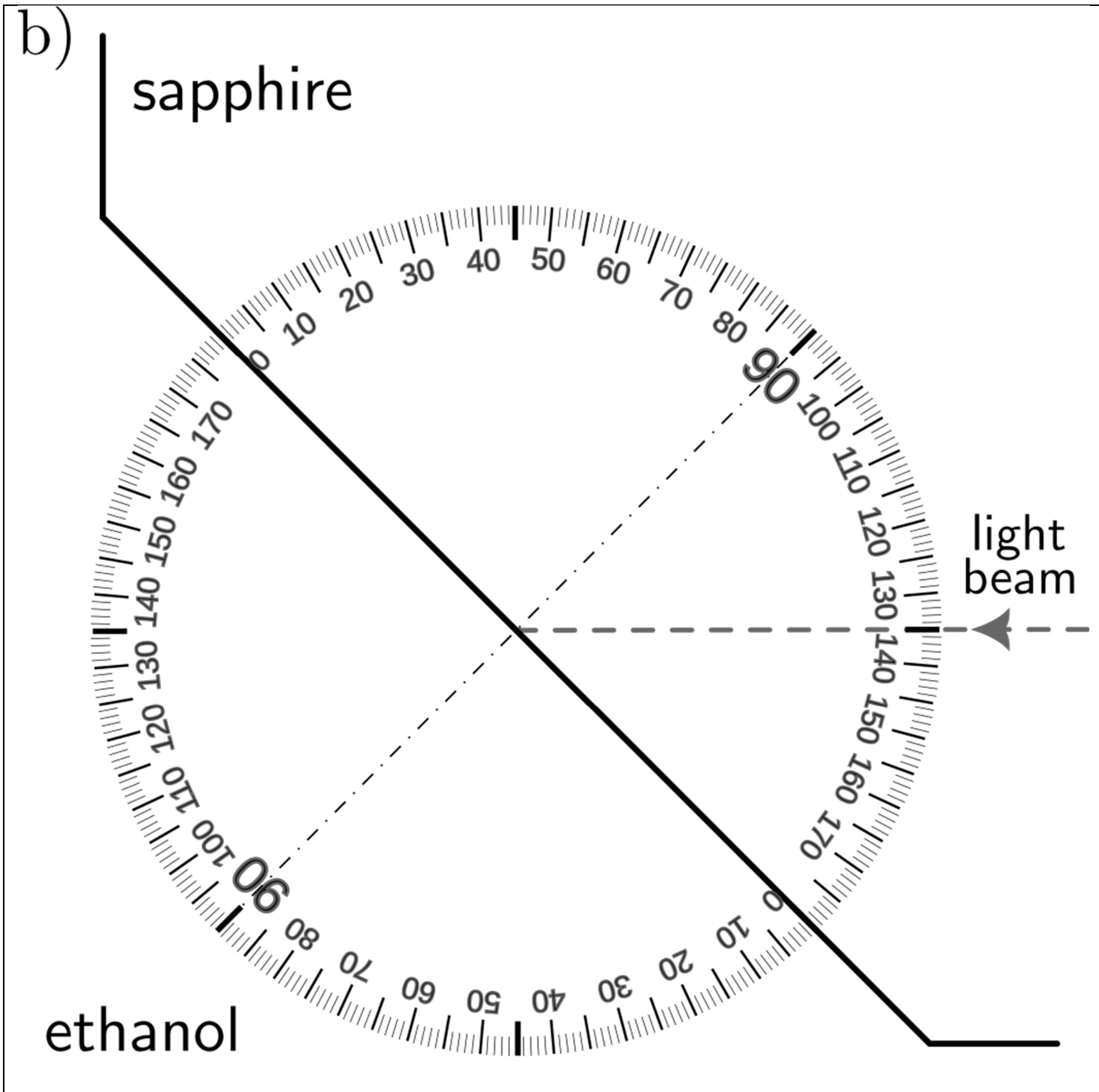
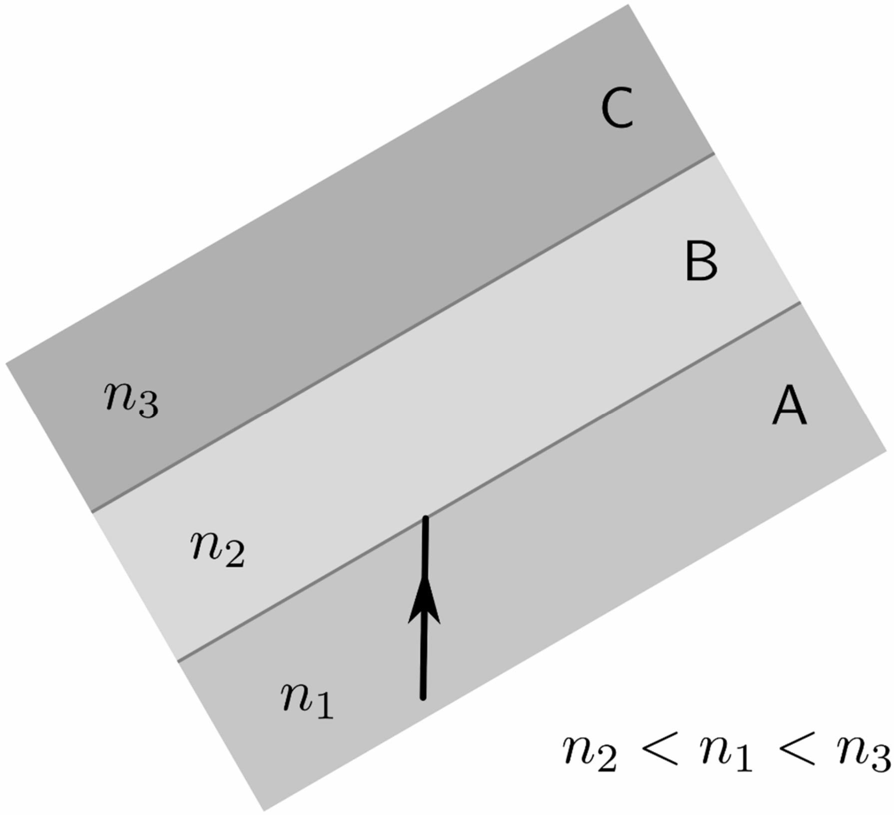
b)	Marks
	
Calculation	

Figure 3.2. Full black line: the boundary between ethanol and sapphire. Gray dashed line: the incoming light beam (the arrow indicates the direction of the propagation). b) The beam reaches the boundary from sapphire.

3.1.2. In Figure 3.3, a ray of light is successively refracted at several boundaries of different media. For the incoming ray in medium A, as indicated in the figure, qualitatively sketch the approximate path of the light ray in media B and C. Pay special attention to the relative direction of the rays in media A and C. (3 p)



Marks

Figure 3.3. The black line represents the direction of the incoming light ray in medium A (with refractive index n_1).

3.2.1. (1 p), *3.2.2.* (3 p), *3.2.3.* (9 p) *and 3.2.5.* (9 p)

Table 3.3. The table for entering the measured and the calculated values.

α(deg)	x exact (cm)	y exact (cm)	x(cm)	y(cm)	d(cm) 0%	d(cm) 20%
0						
5						
10						
15						
20						
25						
30						
35						
40						
45						
50						
55						
60						
65						
70						
75						
						Marks

(Note: if it is not possible to perform all the measurements, the last row of the table may be left empty.)

3.2.4. Add the appropriate amount of table salt to the tub so that you get a 20 % mass fraction aqueous solution. (The '20 % mass fraction' means: for 24 g of demineralized water add 6 g of salt.) Write the mass and height of demineralized water, and mass of added salt. (1 p)

Write the mass and height of demineralized water, and mass of added salt in the Answer Sheet.	Marks
Mass of water:	
Height of water:	
Mass of added salt:	
TOTAL MARKS	

3.3.1. Using a linear fit, determine the slope of $Y = Y(X)$, and hence the refractive indices for demineralized water and the aqueous table salt solution. Write the obtained values (together with their errors with two significant digits). (4 p)

		Marks
Demineralized water	$n =$ ____________ $\pm$ ____________	
20% aqueous NaCl solution	$n =$ ____________ $\pm$ ____________	
TOTAL MARKS		

3.3.2. In the experiment and the calculation, we ignored the influence of the plexiglass tub walls. Based on the value of the refractive index of plexiglass from Table 3.1, estimate whether the true values of the refractive indices of the liquids are ***higher***, ***the same,*** *or* ***lower*** *than the ones measured in the experiment. Check the answer (☒) in the table below.* (2 p)

			Marks
True values of the refractive indices of the liquids are	☐ ***larger***	than the measured values of the refractive indices of the liquids	
	☐ ***the same***		
	☐ ***smaller***		